

# Multivariate normal and Bayesian linear regression

Condition a Gaussian on what you see and a Gaussian comes back. With a Gaussian prior on the weights, the posterior is exact.

## THE QUESTION

Instead of one best weight vector, can we keep all that fit, each weighted by how plausible it is, and predict with them?

## THE INTUITION

Put a Gaussian prior on the weights. With Gaussian noise, log prior plus log-likelihood is quadratic in  $\theta$ , so the posterior is Gaussian. Its precision is the prior's plus  $\sigma^{-2} X^T X$ .

## KEY TAKEAWAY

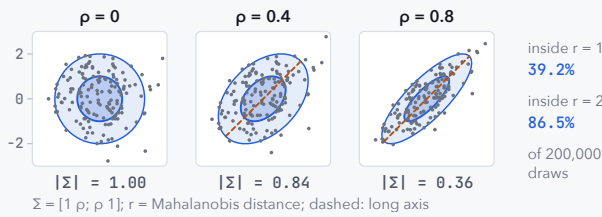
$$S_N = (S_0^{-1} + \sigma^{-2} X^T X)^{-1}$$

$$m_N = S_N (S_0^{-1} m_0 + \sigma^{-2} X^T y)$$

Predictive variance:  $\sigma^2 + x_*^T S_N x_*$

prior  $N(m_0, S_0)$  → likelihood  $N(X\theta, \sigma^2 I)$  → complete the square → posterior  $N(m_N, S_N)$  → predict:  $\sigma^2 + x^T S_N x$

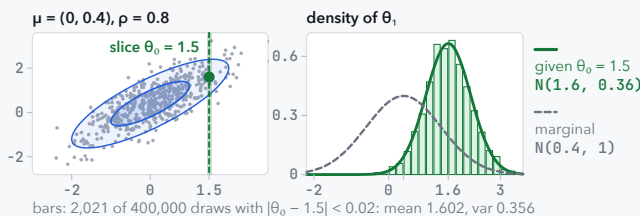
### 1 The covariance matrix sets the ellipse



$$p(\theta) = \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\theta - \mu)^T \Sigma^{-1}(\theta - \mu)\right)$$

Contours: equal Mahalanobis distance  $r$ , the square root of the quadratic form. In 2D  $r^2 \sim \chi_2^2$ , so  $r \leq 1, 2$  hold  $1 - e^{-r^2/2} = 39.3\%, 86.5\%$ . Raising  $\rho$  tilts the ellipse and shrinks  $|\Sigma|$ ;  $\mu$  moves it.

### 2 Marginals drop blocks; conditionals use the Schur complement

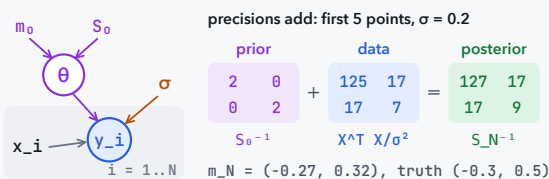


Split  $\theta = (a, b)$  and  $\mu, \Sigma$  into blocks. Marginal:  $p(b) = \mathcal{N}(\mu_b, \Sigma_{bb})$ . Conditional:

$$p(b | a) = \mathcal{N}(\mu_b + \Sigma_{ba} \Sigma_{aa}^{-1} (a - \mu_a), \Sigma_{bb} - \Sigma_{ba} \Sigma_{aa}^{-1} \Sigma_{ab})$$

The covariance (the Schur complement, also  $\Lambda_{bb}^{-1}$  with  $\Lambda = \Sigma^{-1}$ ) ignores the value of  $a$ . Here: mean  $0.4 + 0.8 \cdot 1.5 = 1.6$ , variance  $1 - 0.8^2 = 0.36$ .

### 3 Gaussian prior, Gaussian noise: Gaussian posterior



With the noise variance  $\sigma^2$  known, the posterior is  $\mathcal{N}(m_N, S_N)$ :

$$S_N = (S_0^{-1} + \sigma^{-2} X^T X)^{-1}$$

$$m_N = S_N (S_0^{-1} m_0 + \sigma^{-2} X^T y)$$

Panel 5's first five  $x$  lie in  $[-0.15, 0.43]$ : the slope gets data precision 6.7, the intercept 125.

### 4 Complete the square to read off $m_N$ and $S_N$



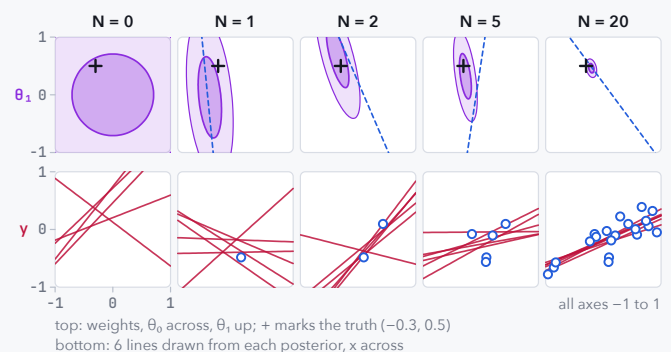
Log-likelihood plus log prior, without the terms free of  $\theta$ :

$$-\frac{1}{2} \left[ \theta^T (\sigma^{-2} X^T X + S_0^{-1}) \theta - 2(\sigma^{-2} X^T y + S_0^{-1} m_0)^T \theta \right]$$

Expand  $\log \mathcal{N}(\theta | m_N, S_N) = -\frac{1}{2} [\theta^T S_N^{-1} \theta - 2m_N^T S_N^{-1} \theta] + \text{const}$  and match: the quadratic term gives  $S_N^{-1}$ , the linear term  $S_N^{-1} m_N$ . In 1D the mean is a precision-weighted average:  $(2 \cdot 0 + 6.71 \cdot 0.503) / 8.71 = 0.388$ .

Or condition the joint Gaussian of  $\theta$  and  $y = X\theta + \varepsilon$  on  $y$  (panel 2):  $m_N = m_0 + S_0 X^T K^{-1} (y - X m_0)$ ,  $S_N = S_0 - S_0 X^T K^{-1} X S_0$ ,  $K = X S_0 X^T + \sigma^2 I$ . It matches to  $7 \cdot 10^{-16}$  (Woodbury).  $S_N$  never uses  $y$ .

### 5 Each point updates the posterior, which becomes the next prior

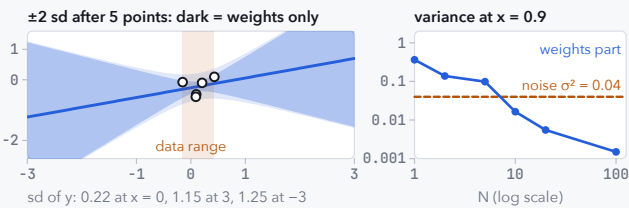


The posterior-figures notebook (after Bishop 2006, Fig. 3.7), replicated draw for draw:  $y = -0.3 + 0.5x + \varepsilon$ ,  $\sigma = 0.2$ , prior  $\mathcal{N}(0, 0.5I)$ . Dashed: weights that fit the newest point exactly, the ridge of its likelihood. The ellipses tilt because the  $x$  values sit right of 0: a steeper line needs a lower intercept (correlation  $-0.50$  after 5 points). Slope sd after 0, 1, 2, 5, 20 points: 0.71, 0.70, 0.53, 0.39, 0.08. After 100,  $m_N = (-0.262, 0.493)$  against the MLE  $(-0.262, 0.494)$ . Point by point equals all at once ( $5 \cdot 10^{-16}$ ).

# Predict with every plausible line

Average the predictions over the posterior: the band holds noise plus weight uncertainty, and it grows where there are no data.

## 6 The predictive adds weight uncertainty to the noise

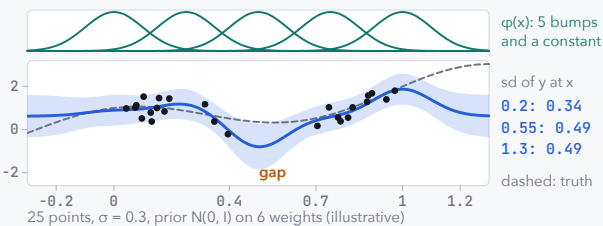


$$p(y_* | x_*, \mathcal{D}) = \int \mathcal{N}(y_* | x_*^T \theta, \sigma^2) \mathcal{N}(\theta | m_N, S_N) d\theta$$

$$= \mathcal{N}(y_* | x_*^T m_N, \sigma^2 + x_*^T S_N x_*)$$

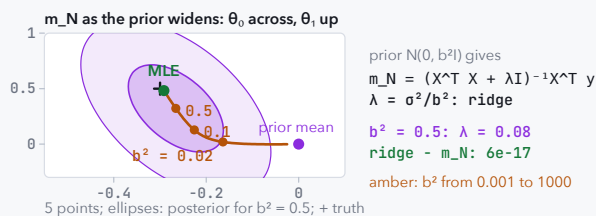
If  $\theta \sim \mathcal{N}(m, S)$  then  $a^T \theta \sim \mathcal{N}(a^T m, a^T S a)$ ; add independent noise.  $\sigma^2$  is aleatoric and stays.  $x_*^T S_N x_*$  is epistemic, least near the data: at  $x = 0.9$  it falls from 0.905 to 0.0015 by  $N = 100$  and is below the noise by  $N = 10$ .

## 7 Basis functions: the same algebra on $\varphi(x)$



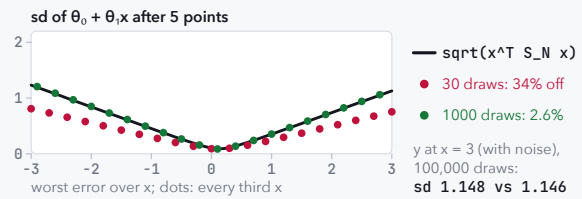
Replace  $x$  by features  $\varphi(x)$ : powers  $x^j$ , sines, or bumps  $\exp(-(x - \mu_j)^2 / (2s^2))$  (the basis notebook: 5 bumps,  $s = 0.1$ ). The model stays linear in  $\theta$ : put  $\Phi$  for  $X$  in every formula. The band widens in the gap. Far from every bump only the constant's prior is left:  $\text{sd } \sqrt{S_{N,00} + \sigma^2} = 0.49$ .

## 8 The MAP estimate is $m_N$ , which is ridge regression



A Gaussian's mode is its mean, so  $\theta_{\text{MAP}} = m_N$ . With  $m_0 = 0$ ,  $S_0 = b^2 I$  it is the slides'  $(X^T X + \sigma^2 / b^2 I)^{-1} X^T y$ : ridge,  $\lambda = \sigma^2 / b^2$  (see the MAP and priors sheet). A tight prior pins  $\theta$  near 0; a flat one gives the MLE, with  $S_N \rightarrow \sigma^2 (X^T X)^{-1}$ .

## 9 Monte Carlo: sample weights, average predictions



Draw  $\theta^{(s)} \sim \mathcal{N}(m_N, S_N)$  and compute  $f_s(x) = x^T \theta^{(s)}$ . Their mean and sd estimate  $x^T m_N$  and  $\sqrt{x^T S_N x}$ , with error  $\propto 1/\sqrt{S}$ ; add  $\varepsilon \sim \mathcal{N}(0, \sigma^2)$  for  $y$ . The MC predictive notebook does this for the prior  $\mathcal{N}(0, I)$  (quoted, 1000 draws): sd 5.23 at  $x = -5$  against  $\sqrt{26} = 5.10$ ; 1.009 against 1.001 near 0. The recipe needs no closed form.

## 10 Bayesian linear regression in NumPy

```
def posterior(X, y, m0, S0, s2):
    P0 = np.linalg.inv(S0)
    SN = np.linalg.inv(P0 + X.T @ X / s2)
    mN = SN @ (P0 @ m0 + X.T @ y / s2)
    return mN, SN
```

```
def predict(Xs, mN, SN, s2):
    var = s2 + (Xs @ SN * Xs).sum(1)
    return Xs @ mN, var
```

```
m, S = m0, S0 # one point at a time
for i in range(len(y)):
    m, S = posterior(X[i:i+1], y[i:i+1], m, S, s2)
```

$(Xs @ SN * Xs).sum(1)$  gives each  $x_*^T S_N x_*$  without the full test covariance. With many features, prefer a Cholesky solve to inv.

## 11 Pause and think

Does the order of the points matter? No: the likelihood is a product.

Can the predictive sd fall below  $\sigma$ ? No:  $x_*^T S_N x_* \geq 0$ .

Does  $S_N$  depend on the targets  $y$ ? No, only on  $X$ ,  $\sigma$  and  $S_0$ .

Where should the next  $x$  go to learn most? Where  $x_*^T S_N x_*$  is largest (the active learning sheet).

What does  $N \rightarrow \infty$  do?  $S_N \rightarrow 0$ ; the band shrinks to  $\pm 2\sigma$ .

Why not predict with the MAP line alone? It drops  $S_N$ : the band would be  $\pm 2\sigma$  everywhere, even far from the data.

Gaussian prior + Gaussian noise  $\rightarrow$  Gaussian posterior: precisions add · predictive variance  $\sigma^2 + x_*^T S_N x_*$  · MAP =  $m_N$  = ridge with  $\lambda = \sigma^2 / b^2$

Source: lecture slides "Multivariate normal distribution" and "Bayesian linear regression" and the notebooks on BLR, posterior figures, basis functions, the Monte Carlo predictive and visualising the normal, nipunbatra.github.io/pml-teaching. Unquoted numbers: NumPy, fixed seeds; panel 5 repeats the notebook's own draw. MVN conditioning, the joint view and the Monte Carlo errors go beyond the slides. Corrections: the posterior-figures notebook swaps its weight-axis labels (across is the intercept); the BLR notebook's "MAP" solves  $(X^T X + \lambda I)\theta = X^T y + \lambda \theta_{\text{MLE}}$ , which returns  $\theta_{\text{MLE}}$ , and its covariance lacks the factor  $\sigma^2 = 0.04$  (band 5 times too wide); the slides' graphical model shades  $x_i^T \theta$  as observed.